Jaejeong Park

AI Autonomy Researcher & Robotics Engineer — Drone Systems · Motion Planning · Vision-Guided Manipulation

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Drone autonomy engineer who has shipped end-to-end autonomous systems — from GPS-denied maritime delivery drones to certified motion planning for eVTOL/UAM (first-author arXiv preprint). 2+ years building perception and navigation stacks with ROS, PX4, and Gazebo in industry; doctoral research at UC Irvine (GPA 3.93) bridging formal verification with real-world aerial robotics. Recently extended this ROS-based systems approach to vision-guided manipulation — leading a multi-robot unmanned-store project and building an IBVS pick-and-place system on ROS 2. Seeking roles where system-level thinking and hands-on robotics experience both matter.

TECHNICAL SKILLS

Languages	Python (Fluent), C++ (Intermediate), MATLAB (Fluent)
Robotics & Simulation	ROS 2 (Jazzy), ROS 1 (Noetic / Melodic), PX4 (MAVROS / MAVSDK), Gazebo (Classic 11, Harmonic), myCobot 280
Perception	Object Detection & Segmentation (YOLOv5/v8, YOLOv8-seg), Multi-Object Tracking (ByteTrack), Visual Odometry (ORB / SIFT), Stereo Vision & 3D Point Cloud (PCL), Pose Estimation (PnP, ArUco / AprilTag)
Planning & Control	MPC (CasADi / IPOPT), RRT*, Image-Based Visual Servoing (IBVS), Jacobian-Based Control, Gaussian Process Regression, Optimal Control
Machine Learning	Certified ML (Monotonic NN), PyTorch, Keras / TensorFlow
Tools	OpenCV, scikit-learn (GPy), Git, Jira / Confluence

EXPERIENCE — ROBOTICS & AUTONOMOUS SYSTEMS

ADDINEDU

Apr 2026 – Jun 2026

Team Lead — Vision-Guided Manipulation using COBOT · *github.com/jaejeongpark/just_pick_it*

- Won the **Grand Prize (1st place)** at the final showcase — judged by industry panels from four companies including LG Electronics and Movensys.
- Led an autonomous unmanned-store project end to end — AMRs transport goods while COBOT arms pick and shelve items on demand.
- Built a vision-guided pick-and-place system on a myCobot 280 collaborative robot with a **ROS 2** architecture.
- Applied **Image-Based Visual Servoing (IBVS)** with Jacobian-based control for manipulation.
- Designed a hybrid closed/open-loop neural network controller (**policy network** + **GripSuccessPredictor**) that learns both the control policy and the grasp timing.

`ROS 2 Jazzy` `myCobot 280` `YOLOv8-seg` `ByteTrack` `IBVS` `Policy Network` `Grasp Success Prediction`

University of California, Irvine

Oct 2023 - Jan 2026

Doctoral Researcher — Electrical Engineering & Computer Science (GPA 3.93/4.0)

- Developed a certified learning-enabled, noise-aware motion planning framework for eVTOL/UAM vehicles — first-author arXiv publication (arXiv:2509.20306).
- Applied Monotonic Neural Networks (MNN) for certified UAM noise prediction — verifiable noise bounds that standard NNs cannot provide — enabling trajectories with guaranteed noise-constraint compliance.
- Integrated RRT* global path planning with certified ML to produce feasible, constraint-satisfying trajectories.
- Teaching Assistant for the graduate Autonomous Systems course: led labs on PID control, path planning, SLAM, and imitation learning.

`Motion Planning` `Certified ML` `Monotonic NN` `RRT*` `eVTOL / UAM` `PyTorch`

Pablo Air Co., Ltd.

Jun 2022 – Jun 2023

Research Engineer — Autonomous Drone Systems (Team PICASSO)

- Led R&D for an end-to-end maritime autonomous drone delivery system operating in GPS-denied open-water environments, integrating detection, navigation, and precision landing into a single ROS + PX4 stack.
- Implemented real-time ship detection using YOLOv5 fine-tuned on custom maritime datasets, combined with CSRT tracker to maintain target lock during occlusion and rapid camera motion.
- Developed monocular visual odometry pipeline (ORB feature tracking + optical flow) for continuous ego-motion estimation over open water where GPS is unavailable; integrated with PX4 EKF2.
- Implemented precision landing on a dynamically moving ship deck using downward-facing camera, YOLOv5 marker detection, and PnP pose estimation (OpenCV solvePnP) to compute 6-DOF relative pose and velocity commands in real time via PX4 offboard control.
- Built full simulation environment in ROS 1 + Gazebo 11: custom camera models, virtual ship model with deck motion, and complete mission pipeline; deployed and tested on PX4-based drone platforms in sea trials.

`Visual Odometry` `PnP / 6-DOF Pose` `PX4 Offboard` `ROS + Gazebo 11` `YOLOv5` `CSRT Tracker`

Dextech Co., Ltd. (joint project with Pablo Air)

Jan 2022 – Jun 2022

Drone Vision Engineer — Structural Inspection & Precision Landing

- Built a fully autonomous structural inspection drone capable of scanning aircraft fuselages without GPS or pre-placed markers: implemented complete pipeline from SIFT keypoint extraction, FLANN matching (Lowe's ratio test + RANSAC), triangulation, PnP 6-DOF pose estimation, to sparse 3D point cloud reconstruction.
- Validated inspection system in Gazebo with high-fidelity aircraft fuselage model; demonstrated robustness of SIFT-based visual odometry across varying lighting conditions.
- Developed autonomous precision landing system for drones operating on moving maritime platforms: YOLO-based landing target detection with custom visual marker, PnP-computed 6-DOF relative pose fed as velocity commands to PX4 offboard controller, compensating for ship deck motion in real time.

`3D Point Cloud (PCL)` `SIFT / FLANN / RANSAC` `Triangulation` `PnP (solvePnP)` `YOLOv5` `PX4` `ROS` `Gazebo`

Hancom InSpace Co., Ltd.

Feb 2021 – Dec 2021

Assistant Researcher — AI & Navigation Systems (Division of Technological Innovation)

- Implemented MPC + Gaussian Process path planning for an autonomous ground robot: kinematic bicycle model, finite-horizon optimal control solved in real time with CasADi/IPOPT, GP (RBF kernel) providing probabilistic path reference with uncertainty bounds.
- Deployed and validated full MPC+GP closed-loop pipeline in Gazebo simulation; result trajectory closely tracked GP reference path even under disturbances.
- Presented MPC path planning research at Korea Aerospace Research Institute (KARI) Future Launcher R&D Program seminar; discussed application to rocket landing control problems.
- Built stereo vision perception pipeline for terrain-relative navigation: camera calibration, stereo rectification, SGBM disparity estimation, 3D point cloud reprojection (PCL), statistical outlier removal, and voxel-grid downsampling — all implemented in ROS.
- Applied MPC+GP path planning to AWS DeepRacer platform for real-hardware validation.

`MPC` `Gaussian Process` `CasADi / IPOPT` `Stereo Vision` `Disparity / PCL` `ROS` `Gazebo` `AWS DeepRacer`

PUBLICATION

Certified Learning-Enabled Noise-Aware Motion Planning for Urban Air Mobility

Jaejeong Park, Mahmoud Elfar, Cody Fleming, Yasser Shoukry · University of California, Irvine & Iowa State University · arXiv:2509.20306 · September 2025

First Author · Preprint

EDUCATION

University of California, Irvine — USA Oct 2023 – Jan 2026
Doctoral Researcher, Electrical Engineering & Computer Science · GPA 3.93 / 4.0

University at Buffalo, The State University of New York — USA Sep 2014 – Dec 2015
Master of Science, Electrical Engineering · GPA 3.83 / 4.0

University at Buffalo, The State University of New York — USA Sep 2010 – Sep 2014
Bachelor of Science, Electrical Engineering · *Magna Cum Laude* · GPA 3.61 / 4.0

PRESENTATIONS

Presenter — MPC Path Planning Seminar KARI Future Launcher R&D Program Office · Dec 2021
Presented self-initiated MPC + Gaussian Process path planning research; led discussion on applying MPC to rocket landing control at Korea Aerospace Research Institute.

PRIOR EXPERIENCE — OPTICAL COMMUNICATIONS (REFERENCE)

The following roles are provided for career timeline completeness. They predate a deliberate self-directed pivot to AI and robotics (2019–2020) and are not the focus of this application.

Licomm Co., Ltd.	Research Engineer — Analog & Digital Depts. SFP optical transceiver design (1.25–6.25 Gbps), R-ONU circuit test & mass production, CE certification.	Feb 2016 – Jul 2019
Fottu Co., Ltd.	Intern — BOSA temperature stress testing; SFP market research.	May 2015 – Jul 2015
Opticis Co., Ltd.	Intern — Fiber optic extension link quality testing; EDID function verification.	Jun 2011 – Aug 2011 / Jun 2014 – Aug 2014